

ORIGINAL ARTICLE

Are thermal constraints shaping foraging ant assemblages in tropical regions? Integrating thermal and morphological traits to understand consequences of land-use transformation

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Abstract

1. Understanding mechanistic processes driving biodiversity change in land-use transformation is paramount but challenging.
2. Functional trait-based approach provides a framework to understand the environmental filtering process of species composition shifts. Yet, trait selection bias may affect the results on forecasting species dynamics, subsequent analyses and the scope of conclusions, in particular, if most studies on terrestrial arthropods rely exclusively on the use of a subset of morphological traits. The integration of ecophysiological traits could then provide additional dimensions of environmental filtering caused by land-use conversion.
3. Our study investigates how land-use conversion from forest to rubber plantation impacts functional traits distribution and trait space in foraging ant assemblages in tropical Thailand by integrating both physiological and morphological traits.
4. The results showed that both physiological and morphological traits may explain, although unevenly, the differences observed in ant assemblages between forest and rubber plantation. The use of Critical Thermal maximum provided an important and unique role for differentiating assemblages encountered in both habitats and in characterising the niche structure of species assemblage from forest.
5. The use of particular traits can thus improve our perception of species dynamic along land-use gradient. To advance our knowledge on the species responses and consequences of land-use changes, the use of different types of functional traits is recommended.

KEYWORDS

biodiversity, Formicidae, functional traits, land-use changes, thermal tolerance

INTRODUCTION

Land-use transformation has been widely recognised as the main driver of global biodiversity decline (Joppa et al., 2016; Newbold

et al., 2015; van Klink et al., 2020). Tropical Asian regions are particularly impacted by anthropogenic conversion associated to rubber and oil palm plantations (Barlow et al., 2018; Gibbs et al., 2010; Hansen et al., 2013), leading to a modification of habitat for native species (Laurance et al., 2014; Warren-Thomas et al., 2015; Wilcove et al., 2013) and monocultures further

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disturbing ecosystem functioning (Barnes et al., 2014, 2017; Ewers et al., 2015; Gossner et al., 2016). Therefore, understanding the consequences of land-use transformation is urgently needed for the conservation of megadiverse tropical regions which harbour most of the planet's biodiversity (Barlow et al., 2018; Basset et al., 2012; Slik et al., 2015).

Species survival in specific habitat relies on their traits such as physiological and morphological aspects (Roeder et al., 2018). Thermal tolerance plays an important role in abiotic filtering, and species tend to be exposed to temperatures close or even exceeding their upper thermal limit in novel habitats following deforestation (De Frenne et al., 2019; Nowakowski et al., 2018; Senior et al., 2018). Because of the canopy cover reduction resulting from deforestation, the crucial role played by vegetation in buffering extreme thermal conditions diminishes (De Frenne et al., 2019; Lee et al., 2021; Scheffers et al., 2014; Senior et al., 2018). Land-use change simplifies the structure of ecosystems and associated microhabitats due to the removal of natural vegetation and disturbances caused by human activities (Barnes et al., 2017; González del Pliego et al., 2016). Species body size, in particular, has been shown to be affected by the levels of habitat complexity and heterogeneity, with simplified habitats favouring smaller body size (i.e., birds: Ehnes, 2019; insects: Gibb et al., 2018). Increasing evidence has suggested that harsh and homogenous habitats (e.g., monoculture, urban habitats and agroecosystems) not only drive species diversity decline (Barnes et al., 2017; Laurance et al., 2014; Wilcove et al., 2013), but also alter species composition and their respective characteristics (Ewers et al., 2015; Karp et al., 2012; Lee et al., 2020; Liu et al., 2016).

Ants are commonly used as a model organism in functional ecology and are frequently regarded as bioindicator of land-use changes (Angilletta Jr. et al., 2007; Lee et al., 2020; Mumme et al., 2015). Using various physiological traits, for example, Critical Thermal maximum (CT_{max}) or thermal tolerance of heat threshold, previous studies have shed light on ant species responses to urbanisation or climate warming (Bujan et al., 2020; Kaspari et al., 2015; Wong et al., 2019). Changes in morphology of ants (e.g., body size) in response to abiotic conditions have also been observed in ant assemblages (Gibb et al., 2018; O'Donnell et al., 2020). Although potential associations between different traits have been suggested, trait selection of functional ecology has mainly focused separately either on the use of thermal traits (e.g., CT_{max}) or on morphological traits (e.g., body size). An integrative approach using both thermal and morphological traits could, thus, provide a more comprehensive understanding in the study of functional ecology and the responses of ant assemblages to land-use transformation.

Here, we pose the following hypotheses: Which set of traits between thermal versus morphological traits are most influenced by land-use change and predominantly contribute to characterising the functional niche of ants? How do thermal traits influence the structure of functional niche and interact with morphological traits? For addressing these hypotheses, we investigated how land-use transformation (from forest to rubber plantation, Figure 1a) shapes

foraging ant assemblages through thermal constraint and habitat simplification in one of the global biodiversity hotspots in Southern Thailand (Myers et al., 2000). Three hypotheses on functional responses of ant species to habitat transformation were tested (Figure 1b–d):

1. *Thermal trait*: Forest with diverse microclimatic niches can harbour ants favouring both wide temperature range so the foraging ant assemblage would present a wider range of low to high CT_{max} values. In contrast, for rubber plantation without dense canopy, diurnal foragers likely experience a hot thermal landscape and ants with low CT_{max} would be excluded (Bates et al., 2013). Thus, we predict that a significant shift and clustering of CT_{max} trait distribution of ants would be observed from forest to rubber plantation with a higher average CT_{max} of ants in rubber plantation than those in forest (Figure 1b).
2. *Occupancy in trait space*: Niche differentiation shapes the functional trait space (FTS) of species assemblage (Salas-Lopez et al., 2018). By mapping species traits on different axes within the FTS (Blonder et al., 2014; Hutchinson, 1957), the spatial changes of species assemblages in trait space can reveal the difference of ecological niches between different habitats. Therefore, combining thermal and morphological traits to quantify FTS of ant assemblages of both habitats, we predict that the trait space occupancy in rubber plantation should be smaller than in forest ant-assemblage due to the functional homogeneity in rubber plantation. Additionally, we expect the thermal trait to play a key role in the FTS due to hotter surface temperature observed in rubber plantation than in forest (Figure 1c).
3. *Niche structure*: Niche structure represents how niche partitioning of species is driven by different functional dimensions within the assemblage. For the specific habitat, rubber plantation presents hotter thermal condition than forest and species would homogeneously present higher thermal tolerance. Therefore, the niche partitioning of ant species within rubber plantation should be driven by factors such as resource acquisition and habitat complexity instead of thermal constraints. We expect morphological traits to play a more unique role in niche partitioning of species assemblage in rubber plantation than forest, while CT_{max} should play a more independent role in quantifying FTS in forest ant assemblage (Figure 1d).

MATERIALS AND METHODS

Study area

Sampling was conducted in Trang province, southern Thailand (7.20–7.60°N, 99.65–99.85°E; Figure S1a,b) during the dry season from 23rd December 2018 to 9th January 2019 (time from 9:00 AM to 4:00 PM). The Tropical Monsoon climate of Trang is under the influence of northeast monsoon from mid-October to

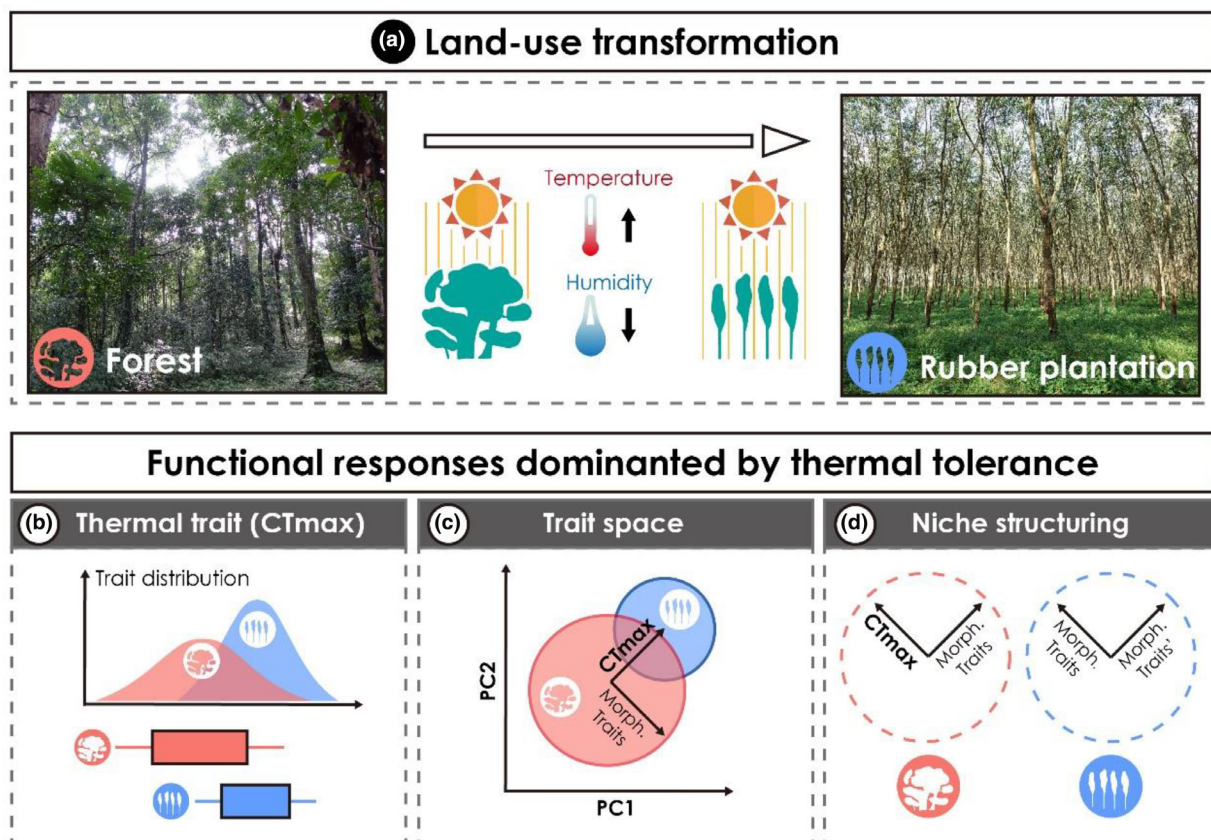


FIGURE 1 Diagram of the predictions. (a) The habitat characteristics changes in land-use transformation. The functional responses are hypothesised to be dominated by thermal tolerance in three facets: (b) thermal trait, that is, the trait distribution and mean value of critical thermal maximum (CT_{max}) in species level; (c) the occupancy of two habitats in trait space and (d) relative roles of traits in niche structuring in two habitats. The trait space is synthesised by both CT_{max} and morphological (Morph.) traits in principal component analysis (PCA) and only the first two components (PC1 and PC2) are shown.

mid-February (Khedari et al., 2002). Under the prevailing monsoon, average temperature of Trang in the dry season is around 26–27°C with around 464.6 mm total precipitation received from November 2018 to March 2019 (Thai Meteorological Department 2019).

Ants were sampled from two habitats, that is, secondary forest and rubber plantation. To sample species assemblage for each habitat and compare species trait responses to land use change, field sites were chosen within representative secondary forests and rubber plantations within the Trang Province (Figure S1). The secondary forest sites are located within the Khao Chong Wildlife Development and Conservation Promotion Station (7.55°N, 99.77°E). Tropical evergreen forests in the region were converted from primary to secondary forests under the action of selective logging activities conducted several decades ago; with current secondary forest recovering in the past 30 years (Personal communications with the station director, Mr. Janya Jareanrattawong). Rubber plantations (7.27°N, 99.66°E) cover large portions of the plain areas in the southern Trang province. We selected several well-managed rubber farms in which understory vegetation and leaf litter are removed annually.

Ants sampling and environmental variables

Graphic data collection and analysis protocol are shown in Figure S2. To collect a sufficient number of foraging individuals per species from both rubber plantations and forests to measure their morphological and thermal traits, baiting method was used as it allows to attract large numbers of alive individuals (Roeder et al., 2018) and accounts for the taxonomic diversity encountered to perform species assemblage comparison (Andersen et al., 2002; Brassard et al., 2021). While the collection of all ant species present within a particular tropical habitat requires tremendous sampling efforts, such holistic approach is not suitable for the sampling of alive individuals to be used in thermal limits assays (Agosti et al., 2000). As a result, sampling was limited to the use of baiting methods; however, to collect a wide range of taxa, and considered for the higher species diversity encountered in forests than in rubber plantations (Liu et al., 2016). We surveyed 10 and 22 sites (i.e., 150 and 330 baiting stations) in the rubber plantations and forests, respectively, representing sufficient sampling efforts for each habitat compared to other studies sampled in tropical regions (Lee et al., 2020; Liu et al., 2016; Salas-Lopez et al., 2018). At each site, 15 baiting stations were randomly deployed for a period of 45–60 min with each baiting station separated by at least five

meters to limit exploitation by a single nest of ants (Eguchi et al., 2004). Each baiting station was composed of 50-ml Falcon tube loosely stuffed with pieces of paper towel to trap foraging workers at the opening of the tube. A 4-mm-thick slice of sausage soaked with honey was deposited at the bottom of the tube and used for collecting foraging ants, as it can provide proteins, lipids and carbohydrates for ant attraction (Andersen et al., 2002). For environmental parameters at each baiting station, ground surface temperature was measured using Fluke 62 Max+ Infrared Thermometer with three replicates around baiting stations; with measurements taken at a distance between the thermometer and ground surface of about 1 m; atmospheric temperature and humidity were recorded by PYLE PTHM 15 m; canopy openness was measured by using Model-A spherical densitometer (Forestry Suppliers, Jackson, MS, USA). The depth of leaf litter is the height of topsoil to the top of litter layer and represents the complexity of ground habitats. This data were obtained in a parallel sampling in the same areas which measured the mean depth of four corners in 1 m² quadrats. Quadrats were located in the corners of each plot (20 m × 20 m). In total, 25 plots were sampled in forest, 14 plots in rubber plantation with the minimum distance of 200 m.

Measurement of physiological response on upper thermal limit

Protocol for measuring the CT_{max} of the different ant species was modified after Diamond et al. (2012), with CT_{max} values determined by the loss of muscle control which represents the most ecologically relevant endpoint (Huey et al., 2012; Lutterschmidt & Hutchison, 1997). Each individual worker collected from the baits was placed in a 1.5-ml Eppendorf tube with clay plugged in the cap to prevent ants from hiding in the coolest space of the tube (Bujan & Kaspari, 2017). Ants were first acclimated indoors at 25–27°C as control temperature for 2 h before the CT_{max} assays. Eppendorf tubes were placed in a digital dry bath (BSH1004), and set at an initial temperature of 36°C. Since the ramping or heating rate influenced the CT_{max} values, a constant ramping rate of 1°C min⁻¹ was maintained throughout the assay. This ramping rate was commonly used for measuring ant CT_{max} and provides the most ecological relevant measurement for ants (used >20 studies, see Leong et al., 2022). At the end of every 1-min interval, each tube was rotated to check whether the ant lost muscle control or not. At least 10 (and up to 15) individuals per species from each habitat were tested for CT_{max}. Two species, *Diacamma* sp. *rugosum* group and *Camponotus* sp. 2, for which fewer than 10 individuals were collected, were only included when using species average but not in modelling intraspecific variation nor trait distribution. After the assays, the ants were preserved in 95% ethanol and point dry-mounted for further identification and measurements of morphological traits.

Species identification and morphological trait measurements

Species-level identifications were first conducted at the genus-level (Eguchi et al., 2011, 2014) and then using relevant Thai ant fauna

checklist (Khachonpisitsak et al., 2020) and available taxonomic keys for the region (i.e., Jaitrong, 2015; Tokeeree & Jaitrong, 2021; Yamane, 2009) to the species level or to morphospecies for undescribed/uncertain species. Overall, the specimens collected were identified to the species-level whenever taxonomic information was available otherwise genus-level and species code were assigned to morphospecies. Then, high-resolution images were taken with a Leica M165c stereo microscope and the software Leica LAS V 4.0 was used to measure the different morphological traits (i.e., WL Weber's Length, HW Head Width, HL Head Length, EW Eye Width, ML Mandible Length, SL Scape Length and FL hind leg Femur Length; see details in Tables S1 and S2). The morphological measurements were taken on at least three individual workers or minors for each species (Gaudard et al., 2019). Finally, for analyses, we used the relative size of five morphological traits scaled to their Weber length (REW Relative Eye Width, REP Relative Eye Position, RFL Relative hind leg Femur Length, RSL Relative Scape Length and RMS Relative Mandible Size). The voucher specimens were deposited at the Insect Biodiversity and Biogeography Lab (IBBL), School of Biological Sciences, University of Hong Kong, Hong Kong SAR.

Functional indices of Critical Thermal maximum

Intraspecific phenotypic plasticity of CT_{max} has previously been demonstrated (Bujan et al., 2020; Gunderson & Stillman, 2015; Sgrò et al., 2016), and intraspecific variation in species' CT_{max} comparison has often been neglected (Franken et al., 2018). For each ant species, we applied a recently developed statistical tool, trait probability density (TPD), to incorporate intraspecific variation and model trait probability distribution (Carmona et al., 2019) which could enhance our understanding of CT_{max} responses to temperature changes. The functional indices, that is, Functional Richness (FRic) and Functional Evenness (FEve) of a trait belonging to each species, were calculated by using TPD (Carmona et al., 2019). We scaled up the species-level TPD to the who assemblage-level for each habitat with equal-weighted species abundance and calculated the assemblage dissimilarity between both habitats. Finally, we also tested the relationship between functional indices and average of CT_{max}.

Functional trait space

We used principal component analysis (PCA) to quantify the FTS of ants. We log-transformed and standardised the mean values of CT_{max} and the six morphological traits (corrected by body size) measured for each species to reduce heteroscedasticity and scale the value to zero mean and unit variance. Next, we excluded the highly correlated morphological traits (pairwise correlation coefficient > 0.5) based on Pearson correlation to avoid unnecessary redundancy, because of unbalanced numbers of traits (i.e., a thermal trait and seven morphological traits). Finally, we produced three FTSs: the first one included all foraging species and investigated the niche occupancy of species assemblages from two habitats within the same regional species pool;

the second and third trait spaces quantified foraging species in forest and rubber plantation, respectively, and investigated the niche partitioning of species within each habitat.

Statistical analyses

The environmental variables, the average and functional indices of CT_{max} in two habitats were compared by nonparametric Mann–Whitney U test. The difference of species assemblages from two habitats in FTS was tested by Permutational Multivariate Analysis using Euclidian distance matrices. The effect of land-use transformation on CT_{max} at individual level was tested by ordinary least squares (OLS) models for the same species collected from both habitats and by linear mixed effect (LME) model for all individuals with species identity set as a random effect. The effect of land-use on functional traits at species level and trait space at assemblage level was also tested by OLS models (see Table S3).

Software

All analyses were applied in R software version 3.4.3 (R Core Team, 2018). R packages used: ‘vegan’ (Oksanen et al., 2013) package for data standardisation, nonparametric Mann–Whitney U test and Permutational Multivariate Analysis; ‘TPD’ (Carmona et al., 2019) for modelling trait probability density and calculating functional indices; ‘FactoMineR’ (Husson et al., 2014) and ‘factoextra’ (Kassambara & Mundt, 2017) for analysing and visualising PCA; ‘nlme’ (Pinheiro et al., 2015) for testing in LME model; ‘ggplot2’ (Wickham, 2016) for producing figures.

RESULTS

Ambient environments of forest and rubber plantation habitats

Temperatures in forest habitat (Figure S1e,f: mean $\pm$ SD, atmosphere = $28.74 \pm 1.60^\circ\text{C}$; ground = $26.00 \pm 0.88^\circ\text{C}$) are 3.73°C and 4.03°C lower on average than within rubber plantation for atmospheric and ground surface temperatures respectively (Figure S1e,f: mean $\pm$ SD, atmosphere = $32.47 \pm 1.72^\circ\text{C}$; ground = $30.03 \pm 2.49^\circ\text{C}$). Conversely, humidity is around 20% lower in rubber plantation (Figure S1g: mean $\pm$ SD, $66.23 \pm 5.21\%$) than in forest habitat (Figure S1g: mean $\pm$ SD, $86.71 \pm 6.35\%$); and canopy openness in rubber plantation is almost 2.6 times larger (Figure S1h: mean $\pm$ SD, $58.61 \pm 12.30\%$) than in forest (Figure S1h: mean $\pm$ SD, $22.72 \pm 17.92\%$). For leaf litter, the depth in forest (Figure S1i: mean $\pm$ SD, 1.29 ± 0.53 cm) is 2.3 times higher than in rubber plantation (Figure S1i: mean $\pm$ SD, 0.58 ± 0.34 cm).

CT_{max} of ants in forest and rubber plantation habitats

In total, 276 individuals of 19 species in forest sites and 127 individuals of 11 species in rubber plantation sites were collected to conduct CT_{max}

measurements. Ant species sampled in rubber plantation displayed significantly higher heat tolerances (mean $\pm$ SD = $48.09 \pm 3.36^\circ\text{C}$) than in forest (mean $\pm$ SD = $45.80 \pm 2.16^\circ\text{C}$), with an average difference of 2.3°C between both species assemblages ($U = 94$, $p = 0.0247$, $n = 30$; Figure 2). The Trait Probability Distribution of CT_{max} at assemblage-level showed that for most ant species CT_{max} values ranged from 41°C to 51°C .

Only two species with individuals collected from multiple sites for each species were collected in both habitats: *Anoplolepis gracilipes* and *Odontoponera denticulata*, allowing for intraspecific comparison in their thermal tolerance between populations retrieved in forests and rubber plantations. Forest populations of *A. gracilipes* showed a significant lower average CT_{max} value by 0.72°C than conspecific workers collected in rubber plantations (Figure S3a: mean $\pm$ SD CT_{max} [forest] = $48.60 \pm 0.95^\circ\text{C}$, $U = 90.5$, $p = 0.029$, $n = 34$); forest populations of *O. denticulata* also showed a similar trend with lower average CT_{max} value by 1.10°C than rubber populations (Figure S3b: mean $\pm$ SD CT_{max} [rubber plantation] = $47.60 \pm 0.66^\circ\text{C}$, $U = 28.5$, $p < 0.001$, $n = 28$).

For intraspecific variation, ants in forest show higher Functional Richness (FRic) and Functional Evenness (FEve) of CT_{max} than in rubber plantation (Figure S4a,b). Moreover, no correlation between functional indices, that is, FRic and FEve and species CT_{max} is observed; at the exception of FEve in rubber plantation assemblage which shows a significant positive correlation with species CT_{max} (Figure S4c,d).

Occupancy of two assemblages in FTS

The Pearson correlation resulted in five functional traits showing weak correlations (i.e., coefficient value: 0–0.5) at species level (Figure S5): Weber’s length (WL), relative mandible size (RMS), CT_{max} , relative scape length (RSL) and relative eye position (REP).

For the functional space including species from both habitats (FTS1), the first three components explained over 80% of the variance in the PCA (Table S4 and Figure S7). The first component (accounted for 39.6% of variance) mainly represented by body size (i.e., WL) and REP and the second component (27.5% of variance) was mainly represented by CT_{max} and RMS (Figure 3 and Table S4). Ant assemblages presented similar occupancy and high overlap on the first two components. The third component (16.1%) predominately represented RSL. On the space quantified by the first and third components, ant assemblage of rubber plantation showed significant smaller occupancy than assemblage from forest ($p = 0.03$). The decrease of FTS occupancy of rubber plantation species was independent to the lower species richness observed there than in forest assemblage (Figure S8).

Effects of land-use transformation on traits and trait space

Regression models suggested that the increase of CT_{max} of ants from forest to rubber plantation could be significantly explained by land-use conversion at both individual and species levels (Table 1).

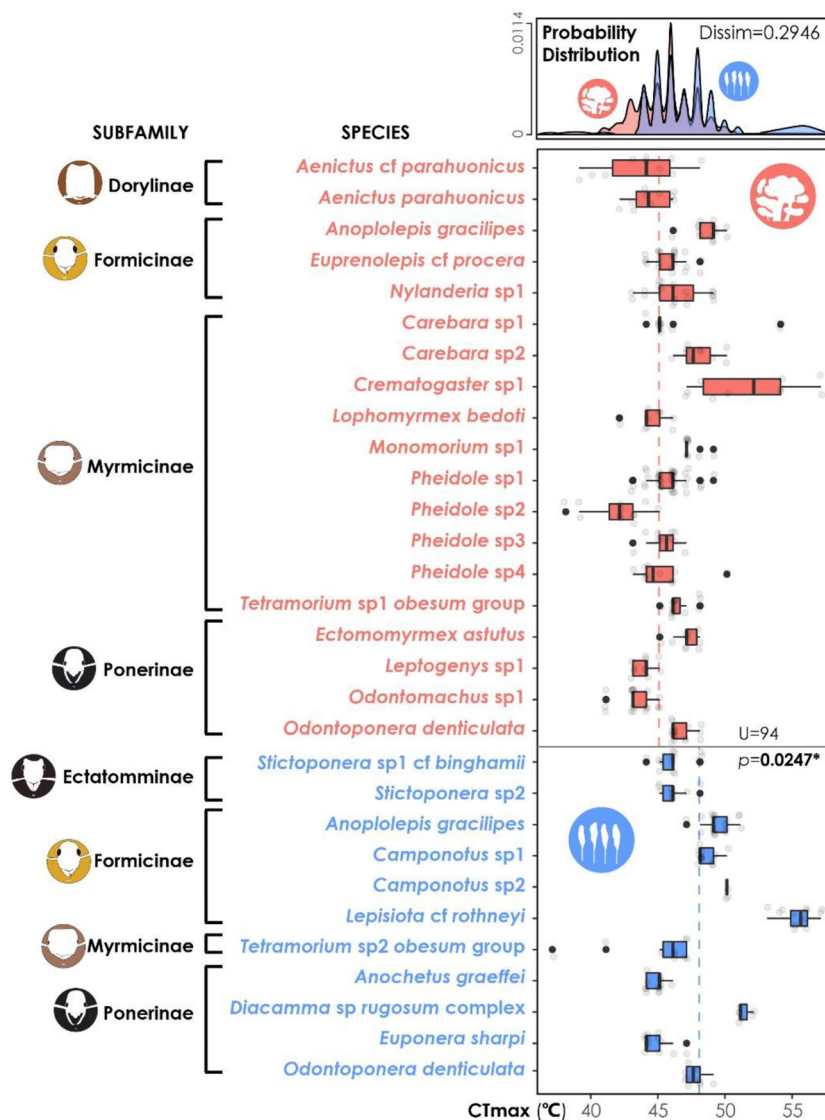


FIGURE 2 Top part: Trait probability density (TPD) incorporating intraspecific variation of critical thermal maximum (CT_{max}) in assemblage-level with presenting the TPD's dissimilarity (Dissim), and the purple colour represents the overlapped TPD between the forest and rubber plantation ant species. Lower part: CT_{max} of ant species from forest (red section) and rubber plantation (blue section) with p value of nonparametric Mann–Whitney U test of species average CT_{max} difference between the two habitats, and the dash lines indicate the mean value of CT_{max} across species in each habitat.

The functional decrease of species assemblage from rubber plantation on the third dimension of functional space was also influenced by the land-use transformation ($p < 0.01$, $R^2 = 0.23$; Table 1).

Niche structuring within two habitats

The niche structuring of FTS within each habitat is quite distinct (Figure 4). For ant assemblage in forest (FTS2), CT_{max} is relatively independent to all other traits and mainly represented by the second (23.0% of total variance explained) and third (16.3% of variance) components (Figure 4a,b and Table S4). However, for communities in rubber plantation (FTS3), CT_{max} is highly correlated with the REP and RSL (Figure 4c,d), and well represented by the first component (representing 45.6% of variance).

DISCUSSION

How land-use transformation drives community composition shift is not fully understood (Barnes et al., 2014, 2017; Ewers et al., 2015). In this study, we integrated thermal and morphological traits to quantify the functional aspects of species assemblage changes in land-use transformation from forest to rubber plantation. Our analyses showed that thermal constraints appear to play key roles in assemblage reorganisation, resulting in higher heat tolerance of ants collected in rubber plantation than in forest. In addition, ant assemblage within rubber plantation sampled here presented a lower functional diversity than in forest.

Habitat conversion modifies the canopy openness resulting in temperature and humidity alteration, potentially leading to a decline of a subset of species with low heat tolerance (Nowakowski et al., 2018;

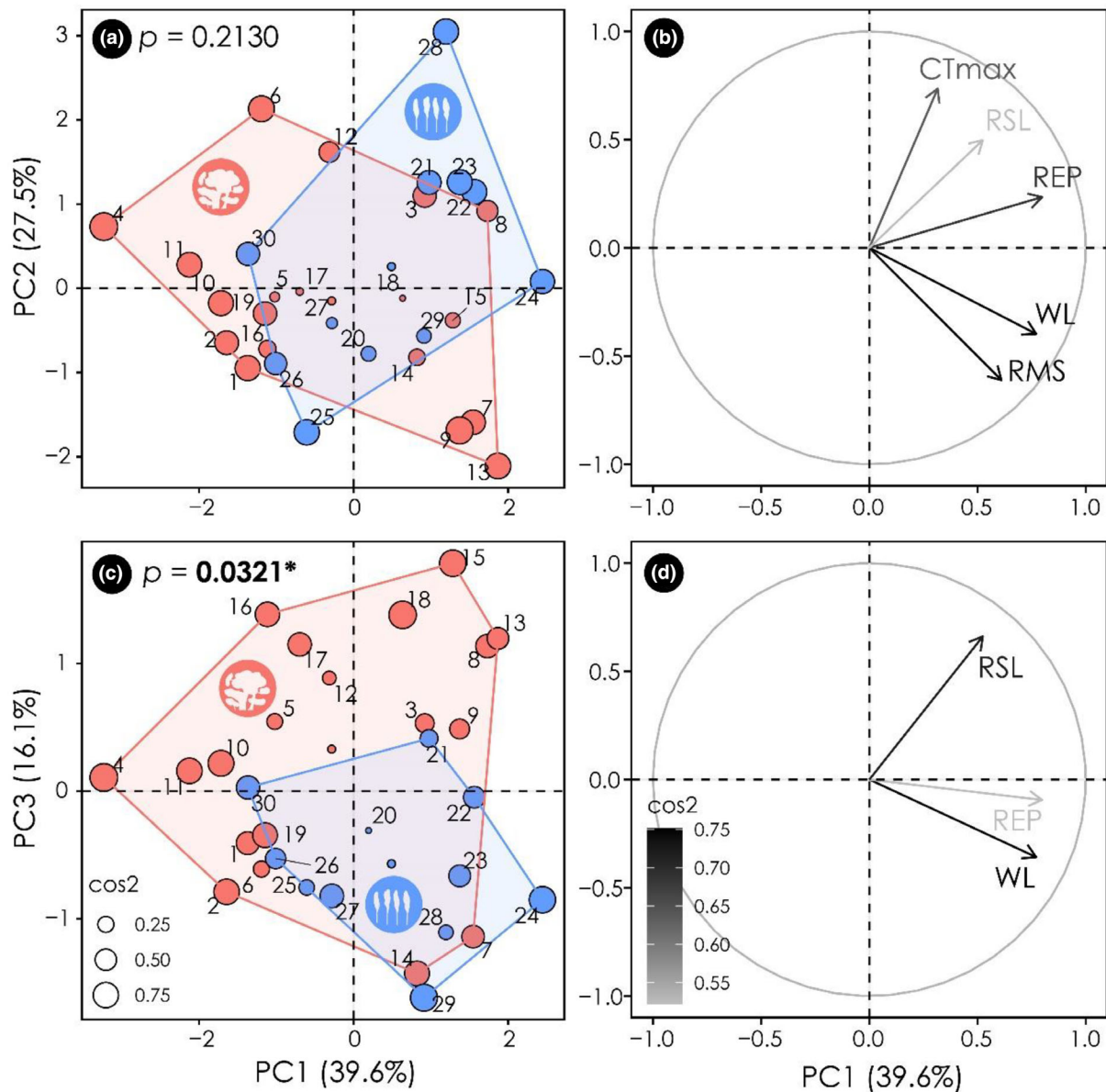


FIGURE 3 Functional trait space (FTS) and niche structure of ant assemblages from the forest and rubber plantation (FTS1) in PCA. (a, b) The first two components mainly represent the habitat-related (WL, Weber's length and RHF, relative hind femur length) and trophic-related (REP, relative eye position and RMS, relative mandible size) morphological traits and Critical Thermal maximum (CT_{max}). (c, d) the third component mainly represents sensory-related (REW, relative eye position and RSL, relative scape length) morphological traits. Each dot in trait space represents a species. Asterisks indicate statistical significance from permutational multivariate analysis of variance between two assemblages (* $p < 0.05$, ns not significant). $\cos^2$: cosine² of parameters in PCA. PCA, principal component analysis.

Tuff et al., 2016). Our results show that foraging ants in rubber plantations present higher thermal tolerance than those in forests (Figures 2 and S3) and a shift in the trait probability distribution of ant assemblages suggesting a loss of ants with lower CT_{max} but a gain of species able to tolerate higher temperature in rubber plantation habitat. This process was also observed in community shift of other ectotherms driven by different types of land-use change (Frishkoff et al., 2015; Nowakowski et al., 2018). Our results also suggest a land-use effect on the thermal tolerance of foraging ants at individual level (Table 1) and the stronger clustering of intraspecific variation of CT_{max} in rubber

plantation than in forest (Figure S4). It should be noted that thermal acclimation may also contribute to the assemblage composition, although it is unlikely to be the major mechanism driving the composition change in our study due to the strong species turnover observed and as observed in previous studies. Ants as small ectotherms are sensitive to local climate, and species CT_{max} is correlated with microclimate (Roeder et al., 2021). For instance, CT_{max} values of Bornean ant species were restricted by habitat temperatures of forest and oil palm plantation, corresponding with the high ant genera turnover (Boyle et al., 2021). Similarly, Mediterranean ant species' habitat specialisation could

TABLE 1 Results of regression models using land-use as the predictor at multiple levels. Ordinary least squares (OLS) models are used for all response variables except critical thermal maximum of all ant individuals ($CT_{\max}$ all), where linear mixed-effect model is used to control the random effect of species identity.

	Estimate	SE	DF	p Value	R ²
<i>Individual-level</i>					
$CT_{\max}$ angr	0.7158	0.3545	32	0.1809	0.1130
$CT_{\max}$ odde	1.0728	0.3045	26	0.0017*	0.3430
$CT_{\max}$ all	1.0626	0.3776	387	0.0051*	
<i>Species-level</i>					
$CT_{\max}$ mean	0.7860	0.3558	28	0.0355*	0.1484
$CT_{\max}$ FRic	-0.6449	0.3996	24	0.1196	0.0979
$CT_{\max}$ FEve	-0.7905	0.3886	24	0.0531	0.1471
<i>Assemblage-level</i>					
FT1 Dist	-0.1755	0.2159	28	0.4230	0.0231
FT1 PC1	0.7744	0.5317	28	0.1560	0.0704
FT1 PC2	0.4084	0.4531	28	0.3750	0.0282
FT1 PC3	-0.8968	0.3081	28	0.0070*	0.2323

Note: Asterisks indicate statistical significance (* $p < 0.05$).

Abbreviations: angr, *Anoplolepis gracilipes*; odde, *Odontoponera denticulata*; FRic, Functional Richness; FEve, Functional Evenness; Dist, PC1, PC2 and PC3, Euclidean Distance and the first three principal components of species assemblage in principle component analysis (PCA).

be predicted by the species $CT_{\max}$ (Arnan & Blüthgen, 2015). Altogether, these studies suggest that habitat temperature represent important constraints on heat tolerance, $CT_{\max}$, among species rather than within species. Thus, species thermal tolerance is most likely to present a limit in species establishment within a particular habitat, especially within those experiencing harsher conditions, than intraspecific variations resulting from acclimation. Collectively, these results here suggest that thermal constraints on foraging species could be a key mechanism in sorting species composition (Frishkoff et al., 2015; Nowakowski et al., 2018).

To adapt to the warmer temperatures and simplified habitat occurring in rubber plantation, ant species present diverse adaptations leading to altered traits composition (Lee et al., 2020; Liu et al., 2016). We found a strong effect of land-use change on the FTS of ant assemblages (Table 1 and Figure 3), with a particular clustering of morphological traits in rubber plantation assemblage which is largely nested within the trait space of forest assemblage (Figure 3c). The decrease of morphological diversity could be partially explained by the decline of species richness (Figure S8). However, the strong species compositional turnover is likely shaped by habitat changes and is consistent with the previous study on foraging ants (Nooten et al., 2019): the monoculture and leaf litter removal of rubber plantation decreases the complexity of habitat and thus excludes small-sized ants (i.e., the size-grain hypothesis; Gibb et al., 2018; Kaspari & Weiser 1999; Figure 3c and Figure S6);

while the hot and high exposure foraging conditions may also constraint the sensory-use (more visual rather than chemical navigation) and result in the clustering of sensory-related traits of species in rubber plantation (Figure 3c).

However, there is an extension and shift in the occupancy of rubber plantation assemblage along the dimension of thermal trait (i.e., $CT_{\max}$) in functional space and its size is similar to the occupancy of forest species (Figure 3a). Abiotic filters of land-use changes have been shown to act on particular physiological and morphological characteristics (Diamond et al., 2018; Gibb et al., 2018), but our result may suggest distinct consequences of different functional aspects responding to the same environmental change. The role that thermal trait plays in ant functional niche partitioning in the rubber plantation is highly correlated to morphological traits but more unique in explaining the ecological niche for forest ant assemblage (Figure 4). Therefore, the integration of thermal trait directly reveals the effect of temperature in shaping the species and functional compositions in habitat conversion. The inclusion of multiple functional aspects (e.g., physiology and morphology) promotes the mechanistical understanding of species reorganisation in environmental changes.

Limitations of this study

In the present study, we found that the integration of multiple traits in structuring niche space is paramount and allows, in part, to explain land-use conversion impacts. Nevertheless, some limitations in our study restrict the generalisation of our findings. First, $CT_{\max}$ characterises the upper thermal limit as one of the aspects of thermal tolerance. We here used it as an example and showed how the inclusion of a physiological trait can contribute to previous knowledge on functional niche characterised by ant morphology in different habitats. But other thermal traits, such as desiccation resistance or radiation dispersal abilities, could also influence the tolerance of species to temperature and humidity (Bujan et al., 2016; Kellermann & van Heerwaarden, 2019; Law et al., 2020; Schowalter et al., 2017). Identifying and integrating additional key ecological traits could further enhance our understanding of ecological processes in the future. Second, our study described the response of ant assemblages occupancy in FTS to habitat conversion; however, species interactions in function of temperature are also important in community ecology (Wittman et al., 2010). Third, ant communities present a variety of microhabitat use and temporal patterns. For instance, arboreal ants foraging in the canopy can experience harsher thermal conditions than ground-dwelling species, while nocturnal ants experience more similar temperature conditions in both forest and rubber plantation (Bestelmeyer et al., 2000; Cerdá & Retana, 1997; Kaspari et al., 2015). Our sampling focused on diurnal ground ant foragers and these results may not fully be representative of other community subsets exposed to different foraging conditions; which would deserve particular studies on their own.

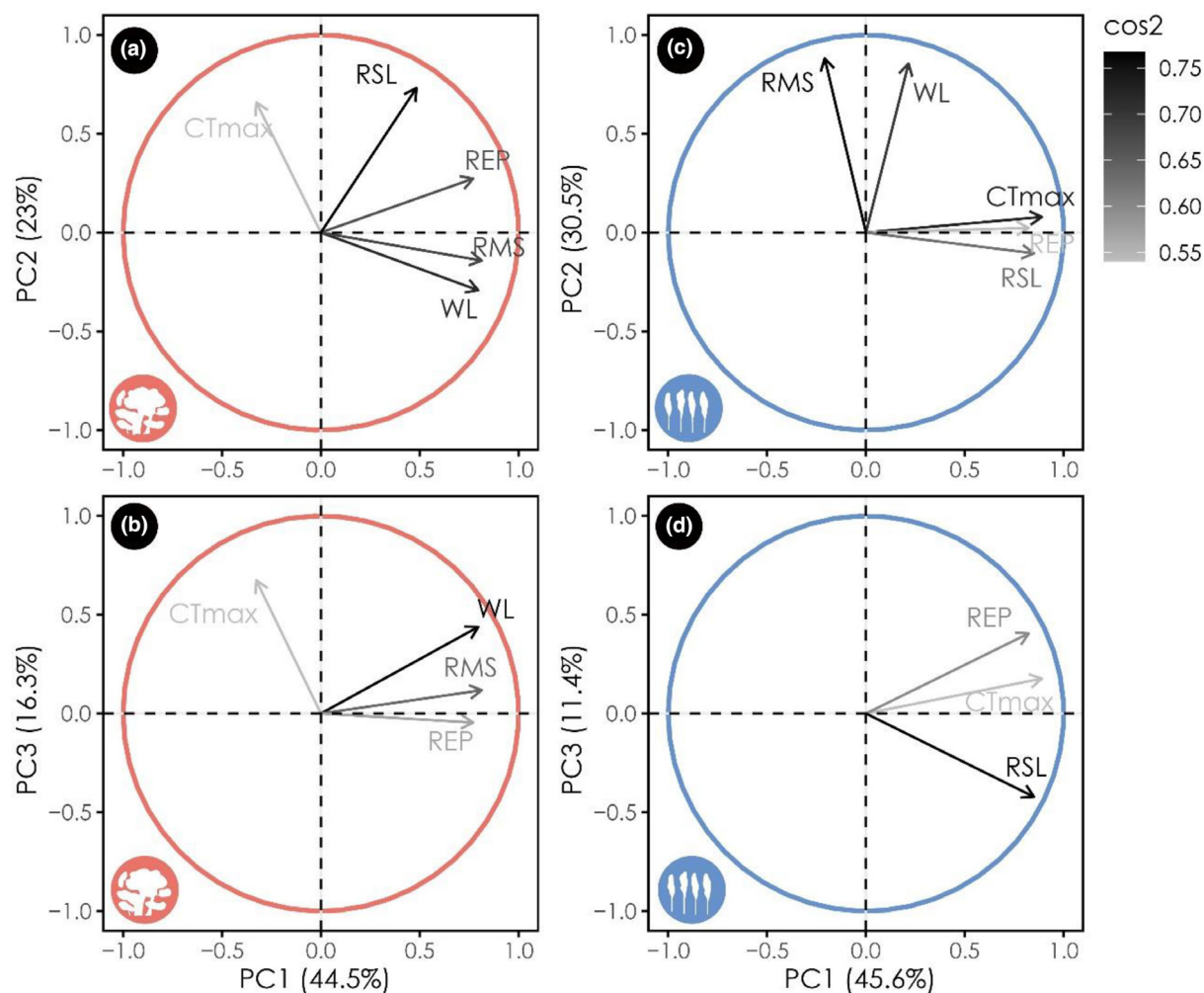


FIGURE 4 Functional trait spaces (FTSs) of the ant species collected from (a, b) forest (FTS2) and (c, d) rubber plantation (FTS3) in PCA. $\cos^2$: cosine² of parameters in PCA. PCA, principal component analysis.

CONCLUSION

Our results demonstrated the significant effects of habitat conversion from forest to rubber plantation on the ant community selection of some physiological and morphological traits, as well as the occupancy of ant assemblages in the FTS. In general, the maximum thermal tolerance of ants tends to be higher and their morphological functional diversity lower when habitats shift from a more complex to a simpler and hotter environment. These results reveal that habitat transformation shapes a subset of the ant species composition through multiple processes including thermal constraint and structural loss. On the other hand, the functional trait-based approach can provide a mechanistic understanding of land-use transformation consequences, especially when integrating physiological and morphological approaches. We found that morphological traits could characterise most of the variations observed in the assemblage FTS, but that the addition of a physiological trait shed a new light on a distinct thermal constraint process. This suggests that thermal constraints may play a key role in driving functional changes through land-use conversion and may

further affect ecosystem functioning. Information on different biological attributes and their integration in functional ecology could deeply improve our understanding on the impacts of anthropogenic activities and provide new insights for conservation.

AUTHOR CONTRIBUTIONS

Chi-Man Leong: Conceptualization (lead); data curation (lead); investigation (equal); methodology (lead); writing – original draft (equal); writing – review and editing (equal). **Runxi Wang:** Conceptualization (lead); formal analysis (lead); investigation (equal); methodology (equal); visualization (lead); writing – original draft (equal); writing – review and editing (equal). **Shun-Chi Cheung:** Investigation (equal); writing – original draft (supporting); writing – review and editing (equal). **Weeyawat Jaitrong:** Investigation (supporting); resources (equal); writing – original draft (supporting); writing – review and editing (equal). **Benoit Guénard:** Conceptualization (equal); funding acquisition (lead); project administration (lead); resources (lead); supervision (lead); validation (lead); writing – original draft (equal); writing – review and editing (equal).

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CONFLICT OF INTEREST

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data supporting the findings of this study are available within the article and its Supporting Information.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

APPENDIX S1. Supporting Information

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